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Cd(II) adsorption from aqueous solution using pristine and activated *Melocanna baccifera* biochar: isotherm and kinetics studies

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Bamboo biochar is effective at adsorbing a wide range of heavy metals and is relatively stable under environmental conditions. Cadmium contamination in aquatic systems poses a significant risk to environmental and human health, creating a need for efficient, low-cost adsorbents for water treatment. In this study, pristine and chemically activated biochars derived from *M. baccifera* bamboo were evaluated for Cd(II) removal from aqueous solution. The objective was to investigate the effect of chemical activation on adsorption performance and to elucidate the adsorption mechanisms using equilibrium and kinetic analyses. Biochar was activated using various concentrations of potassium hydroxide (KOH) and phosphoric acid (H₃PO₄). Among the tested methods, activation with 60% KOH demonstrated the highest adsorption efficiency, leading to its selection for further characterization and adsorption studies. Batch experiments revealed that the adsorption process is significantly influenced by pH, with optimal removal observed at pH 5. The equilibrium data were best represented by the Langmuir isotherm, indicating maximum adsorption capacities of 15.15 mg/g for pristine biochar and 24.39 mg/g for activated biochar. Kinetic analysis indicated that the adsorption kinetics conform to the Pseudo-second-order model, with Intraparticle diffusion and liquid-film diffusion identified as contributing factors in the rate-controlling steps. Fourier-transform infrared (FTIR) spectroscopy analysis showed shifts in the O-H, C=O, and C-O functional groups. These changes suggest that surface complexation and ion exchange are the primary mechanisms involved in the binding of cadmium ions. The results of this study demonstrate that KOH activation significantly improves the physicochemical properties and adsorption performance of bamboo biochar. This research highlights the potential of activated biochar derived from *Melocanna baccifera* as a sustainable and locally available adsorbent for the removal of cadmium from contaminated water. This also supports its potential application in cost-effective water purification systems.

KEYWORDS

adsorption, bamboo, cadmium, carbon, chemical activation

1 Introduction

Cadmium (Cd), a highly toxic heavy metal, is frequently found in wastewater effluents. High levels of Cd(II) are particularly problematic because of their non-biodegradability, persistence, and extreme toxicity, which endanger living organisms and pose a serious threat to human health (Chen et al., 2023; Zheng et al., 2022). Key sources of cadmium in wastewater include metallurgical alloying, ceramics, metal plating, pigment production, phosphate fertilizers, mining, and sewage sludge (Liu et al., 2023). According to the World Health Organization, the acceptable concentration of cadmium in drinking water and industrial wastewater is 0.003 mg/L (Saeed et al., 2022).

Various techniques can be employed to remove heavy metal ions, such as chemical precipitation, adsorption, ion exchange, reverse osmosis, electrodialysis, electrochemical reduction, ultrafiltration, solvent extraction, and evaporation (Kavisri et al., 2023; Madadgar et al., 2023). Adsorption is a commonly used method due to its simplicity, cost-effectiveness, and high efficiency (Alzahrani et al., 2022). Therefore, there is an urgent need to develop environmentally friendly and highly efficient adsorbents for the removal of Cd(II) from wastewater.

Various adsorbents are currently being used to eliminate Cd(II) from contaminated environments. Biochar is gaining attention due to its numerous advantages (Lakshmi et al., 2021; Zhang et al., 2020). It is created through the slow pyrolysis of biomass in low-oxygen conditions, resulting in a large surface area, high cation exchange capacity, and abundant oxygen-containing functional groups. These qualities make biochar effective in reducing the bioavailability and ecological toxicity of heavy metals (Xu et al., 2023; Zhao et al., 2023).

Various activation methods are currently available to improve the sorption capacity of biochars. Chemical activation, in particular, is a prominent method that increases the surface area, cation exchange capacity, and abundance of functional groups (Leng et al., 2021). KOH activation significantly enhances pore size, surface area, and catalytic activity (Herath et al., 2021). Alkaline treatment enlarges micropores to mesopores, which facilitates adsorbate transport and enhances adsorption capacity (Luo et al., 2016). H₃PO₄ activation also enhances adsorption capacity by introducing phosphoric functional groups (PO₄³⁻), which contribute to surface functionalization and enhance adsorption performance (Jawad et al., 2020). Consequently, a comparative analysis of the adsorption capacity between KOH and H₃PO₄ activation was conducted.

Previous studies have reported that bamboo-derived biochar can effectively adsorb a wide range of heavy metals (Chen et al., 2011). Bamboo grows rapidly and abundantly, making bamboo biochar a cost-effective and sustainable alternative to other biomass sources (Lehmann and Joseph, 2009). Moreover, bamboo biochar is relatively stable under environmental conditions, making it suitable for long-term remediation of heavy metals (Mohan et al., 2006).

While various agricultural wastes and bamboo-derived biochars have been studied for heavy metal removal, there is limited research specifically focused on biochar produced from *M. baccifera*. This bamboo species is abundant in Northeast India and is often underutilized, as it contains significant amounts of lignocellulosic residue. The high cellulose and hemicellulose content of *Melocanna*

baccifera indicates strong potential to produce porous carbon materials with numerous oxygen-containing functional groups after chemical activation, which could enhance metal adsorption. However, systematic studies examining the adsorption behavior of Cd(II) using both pristine and chemically activated *M. baccifera* biochar are lacking, and the relationship between the physicochemical changes induced by activation and the adsorption mechanisms remains unclear.

Therefore, the present study aims to investigate the adsorption performance of *M. baccifera* pristine biochar (MPB) and *M. baccifera* activated biochar (MAB) for the removal of Cd(II) from aqueous solution. The adsorption behavior was analyzed using equilibrium and kinetic models, while Fourier-transform infrared spectroscopy (FTIR) was employed to identify surface functional groups and to elucidate the mechanisms of Cd(II) binding. This work seeks to provide insight into the suitability of *M. baccifera* biochar as a low-cost and sustainable adsorbent for wastewater treatment.

2 Experimental

2.1 *Melocanna baccifera* (bamboo) biochar as an adsorbent

The dried culms of *M. baccifera* were collected and thoroughly washed with tap water to remove any dust and impurities. After air-drying, the material was cut into small pieces, approximately 12–14 inches long, and prepared as feedstock for pyrolysis.

Pyrolysis was conducted in a brick kiln under oxygen-limited conditions. This was achieved by restricting the air inlet and maintaining a closed chamber during carbonization. The temperature was gradually increased at a controlled heating rate until the desired pyrolysis temperature (500 °C), was reached, and then held for 4 h (Mohan et al., 2006; Tripathi et al., 2016). Once pyrolysis was complete, the furnace was allowed to cool naturally to room temperature while still restricting air flow to prevent oxidation of the produced biochar. The resulting pristine biochar, designated as MPB, was then ground and sieved to obtain a uniform particle size of 100 mesh (≈149 μm).

2.2 Activation of biochar

KOH (Himedia) and H₃PO₄ (Qualigens) in the concentrations of 20%, 40%, 60%, and 80% were used for chemical activation. Biochar powder with a mass concentration of 40 g/L was exposed to these different concentrations of chemical reagents for 24 h. The mixture was agitated on an orbital shaker (Orbitek, Scigenics Biotech, Chennai, India) at 150 rpm. Afterward, the excess solution was removed, and the biochar was heated to 800 °C for 30 min using a muffle furnace (Bharat Scientific Equipment, India). The activated biochar was then allowed to cool to room temperature. After activation, the product was repeatedly washed with deionized water until neutral pH was reached to remove residual activating agents, and then oven-dried at 50 °C for 12 h. The dried material was ground and sieved (100 mesh) and designated as MAB (*M. baccifera* activated biochar).

For testing, 0.1 g each of MPB and MAB was mixed with a 10 mg/L solution of Cd(II) and shaken for 6 h at $29 \pm 1^\circ\text{C}$ on the shaker. The biochars were then separated from the supernatants by centrifugation at 10,000 rpm for 8 min and the metal concentrations in the solutions were analyzed.

2.3 Adsorbate

All chemicals used in this study were of analytical grade. A stock solution of cadmium (1 g/L) was prepared by dissolving 2.103 g of $\text{Cd}(\text{NO}_3)_2$ (anhydrous) in Milli-Q water. Various concentrations of Cd(II) were obtained by diluting the 1 g/L stock solution. For calibration in the Atomic Absorption Spectrophotometer, a standard Cd(II) solution (1 g/L) from SRL (India) was used. 1N HNO_3 and 1N NaOH were used to adjust the pH of the solution.

2.4 Adsorbent and its characteristics

The samples were gold sputter-coated, and a scanning electron microscope (XL30 ESEM, Philips, United States of America) was used to capture SEM micrographs for analyzing the surface morphology of MPB and MAB.

The surface area and pore characteristics, including average pore volume, pore diameter, and micropore volume, were determined by N_2 adsorption–desorption analysis using Brunauer–Emmett–Teller (BET) analysis. The measurements were performed using a Micromeritics-ASAP 2020 Surface Area and Porosity Analyzer (United States of America). Samples were degassed at 150°C for 6 h under vacuum to remove moisture and volatile impurities.

Additionally, an FTIR spectrometer (JASCO FTIR-5300) was used to analyze the surface functional groups present on MPB and MAB before and after the adsorption of 30 mg/L Cd(II) ions. Samples were dried at 60°C under vacuum for 12 h, cooled in a desiccator, and scanned immediately after N_2 purging of the spectrometer with spectral wave number range between 400 and $4,000\text{ cm}^{-1}$. The spectra were baseline-corrected uniformly. These steps minimize moisture and CO_2 variability.

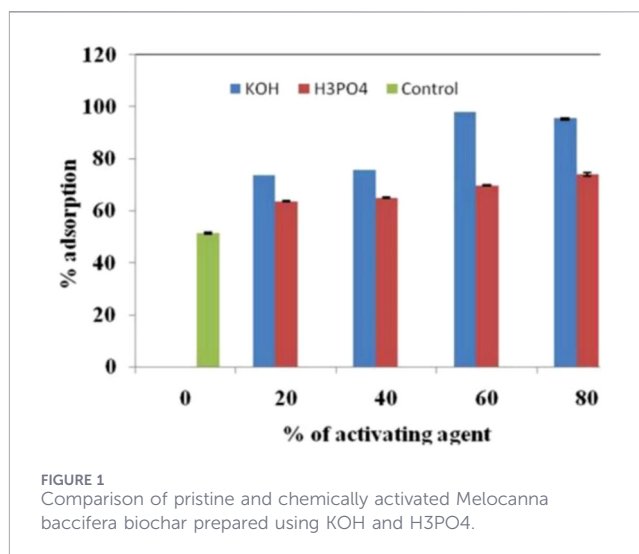
2.5 Batch mode adsorption studies

The adsorption behavior of Cd(II) ions on MPB and MAB was studied using batch adsorption experiments. All experiments were conducted in 250 mL conical flasks, each containing 100 mL of metal solution.

The effect of solution pH was evaluated over a pH range of 2.0–7.0, adjusted using 1N HNO_3 or 1N NaOH. In each experiment, 0.1 g of dried adsorbent was added to the solution, and the mixture was agitated at 150 rpm and $29 \pm 1^\circ\text{C}$ for 6 h.

Additionally, experiments were conducted to study the effect of contact time (from 15–360 min) and initial Cd(II) concentrations (from 5–30 mg/L), while keeping other parameters constant. After adsorption, the suspensions were separated by centrifugation at 10,000 rpm for 8 min, and the residual Cd(II) concentration in the supernatant was determined using a Flame Atomic Absorption Spectrometer (model GBC 932 plus, Australia) at a wavelength of 288 nm.

The instrument was calibrated using a standard solution within a linear calibration range, yielding a correlation coefficient (R^2)



between 0.995 and 1.0. All the experiments were performed in triplicate, and the mean values were used for data analysis. Model fitting was evaluated using goodness-of-fit statistics and comparison between experimental and calculated values. The amount of Cd(II) adsorbed onto biochar at equilibrium, q_e (mg/g), was calculated as follows:

$$q_e = (C_0 - C_e) \times V/W \quad (1)$$

where C_0 and C_e (mg/L) are the initial and equilibrium liquid-phase concentrations of Cd(II), respectively, V is the volume of the solution (L), and W is the weight of the dry sorbent (g).

3 Results and discussion

3.1 Activation of Melocanna baccifera biochar

The adsorption capacity of biochar treated with different concentrations of KOH and H_3PO_4 was evaluated. Biochar activated with 60% KOH exhibited the highest adsorption efficiency, (97.89%) for Cd(II), followed by H_3PO_4 -activated biochar, (73.93%). In comparison, untreated biochar (MPB) showed an adsorption efficiency of 51.39% from an initial Cd(II) concentration of 10 mg/L, as presented in Figure 1. Although the H_3PO_4 activation improved adsorption relative to pristine biochar, KOH activation consistently produced higher removal efficiencies at comparable concentrations. Therefore, biochar activated with 60% KOH treatment was chosen for subsequent experiments.

KOH activation not only enhances porosity but also leads to significant surface chemical modifications. During the carbonization process, KOH reacts with carbon ($6\text{KOH} + 2\text{C} \rightarrow 2\text{K} + 3\text{H}_2 + 2\text{K}_2\text{CO}_3$), producing metallic potassium (K), carbon dioxide (CO_2), and potassium carbonate (K_2CO_3) (Wang and Kaskel, 2012). These products, etch the carbon matrix and create oxygenated edge sites. When these sites are exposed to air, they oxidize, resulting in the formation of $-\text{OH}$, $\text{C}=\text{O}$, and $-\text{COOH}$ groups (Wang et al., 2020).

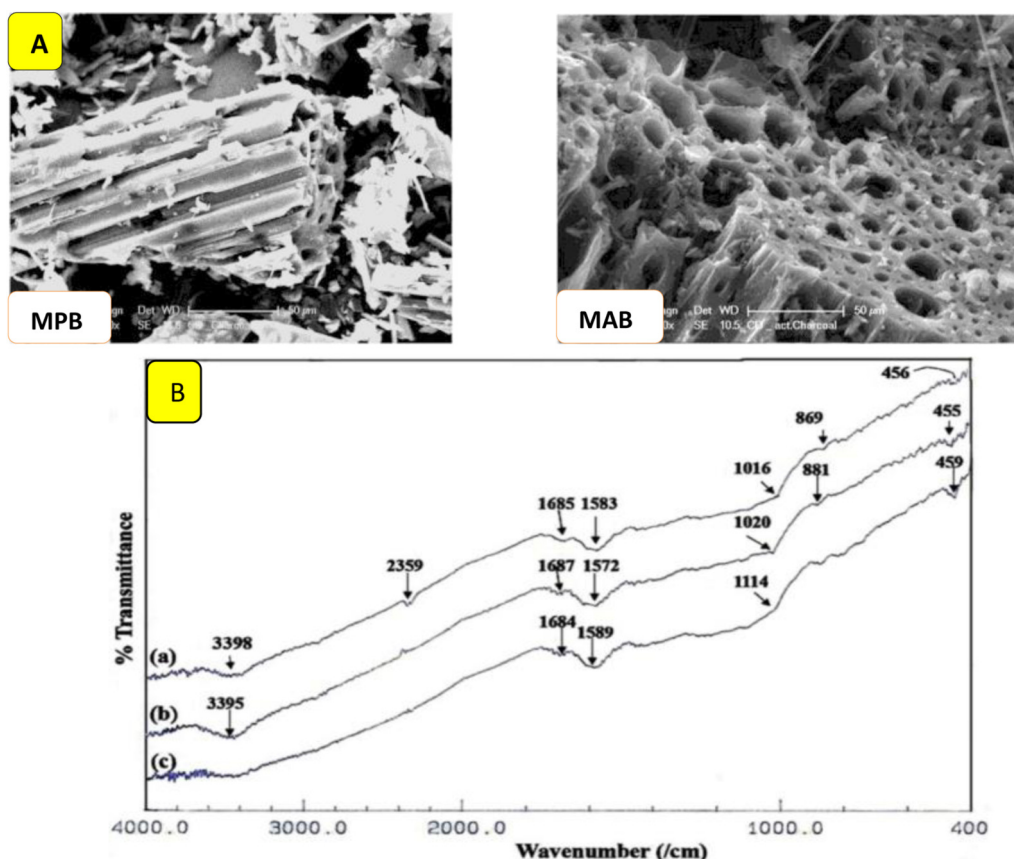


FIGURE 2

(A) Scanning electron microscopy (SEM) micrographs of *Melocanna baccifera* pristine biochar (MPB) and *Melocanna baccifera* activated biochar (MAB), showing changes in surface morphology after chemical activation. (B) Fourier-transform infrared (FTIR) spectra of (a) dried unloaded biochar, (b) Cd(II)-loaded MPB, and (c) Cd(II)-loaded MAB, recorded in the range of 400–4,000 cm^{-1} , illustrating changes in surface functional groups before and after adsorption.

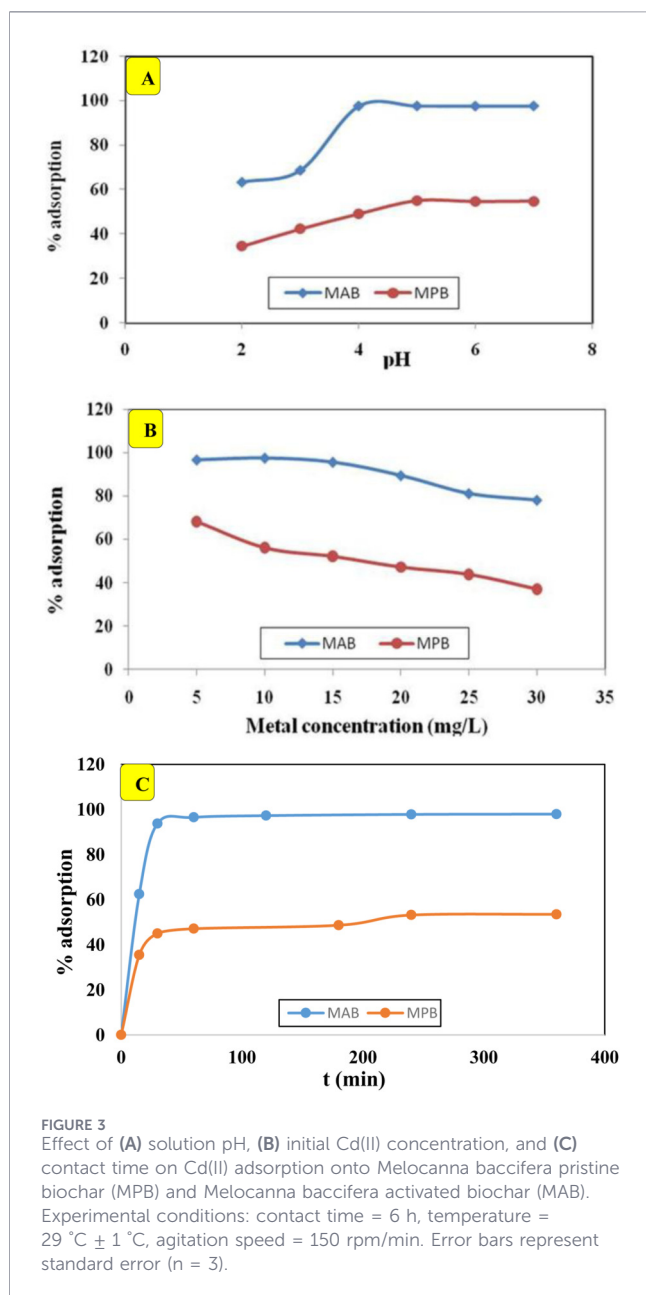
The FTIR spectra of modified activated biochar (MAB) confirm this enrichment, showing intensified or shifted bands at approximately 3,395 cm^{-1} (O–H), 1,684–1,589 cm^{-1} (C=O/C=C), and 1,016–1,114 cm^{-1} (C–O). These functionalities serve as ligands for cadmium ions (Cd(II)) through metal–oxygen coordination and ion exchange. This explains why MAB has a higher adsorption capacity for Cd(II) beyond just the increase in surface area. The combined effects of textural activation and chemical functionalization contribute to the improved uptake of Cd(II) observed in KOH-activated biochar.

3.2 Adsorbent and its characteristics

SEM micrographs of the biochars reveal a porous surface morphology, suggesting increased pore development and availability of adsorption sites. As shown in Figure 2A, the activated biochar (MAB) exhibits more developed pore structures and larger pore openings compared to pristine biochar (MPB). Chemical activation significantly improved the textural properties of the biochar: the surface area increased from 322 m^2/g to 432 m^2/g , the average pore volume rose from 0.154 cm^3/g to 0.212 cm^3/g , and the micropore volume grew from 0.084 cm^3/g to 0.108 cm^3/g .

The IR spectra of unloaded biochar, Cd(II)-loaded MPB, and Cd(II)-loaded MAB were analyzed to investigate the surface functional groups (Figure 2B). The spectra were recorded in the range of 400–4,000 cm^{-1} to identify changes in vibration frequencies of the functional groups involved in metal ion adsorption.

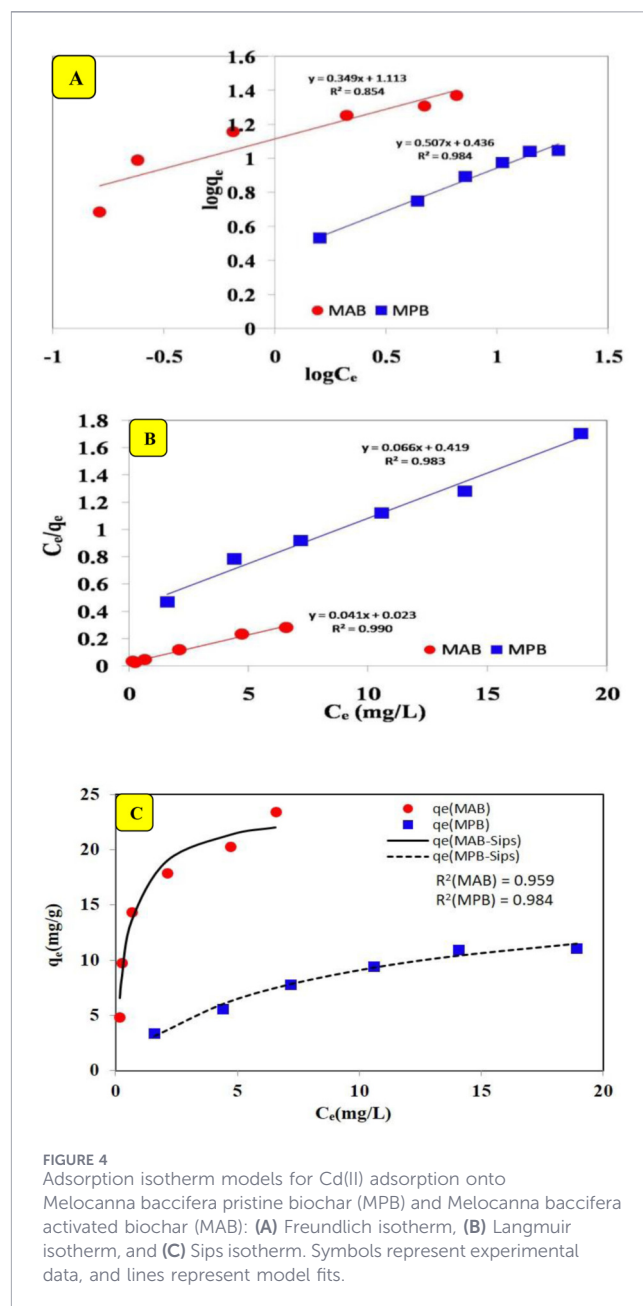
The band observed near 2,359 cm^{-1} is attributed to atmospheric CO_2 present within the optical path and/or weakly adsorbed on the porous surface and is not associated with the functional groups involved in Cd(II) adsorption. The broad envelope around ~3,395 cm^{-1} , which corresponds to O–H/N–H stretching, decreases in intensity and shifts slightly after the loading of Cd(II). This change is consistent with the deprotonation and coordination of surface hydroxyl and phenolic groups, as well as minor variations in residual moisture. The spectral band at 1,684–1,687 cm^{-1} , corresponding to the stretching vibration of carboxyl groups (–C=O), shifted from 1,685 cm^{-1} to 1,687 cm^{-1} for Cd(II)-bound raw biochar and to 1,684 cm^{-1} for Cd(II)-loaded activated biochar. The bands between 1,572 and 1,589 cm^{-1} , associated with C=O stretching vibration, shifted from 1,572 cm^{-1} to 1,583 cm^{-1} and 1,589 cm^{-1} for Cd(II)-adsorbed pristine and activated biochars, respectively. The adsorption band at 1,016–1,114 cm^{-1} , related to C–H bending and C–O



stretching vibrations, also shifted after Cd(II) adsorption. Shifts in C=O/C–O bands ($\approx 1,684$ – $1,687$ and $1,572$ – $1,589\text{ cm}^{-1}$) and in the C–O region ($\approx 1,016$ – $1,114\text{ cm}^{-1}$) indicate inner-sphere complexation of Cd(II) with carboxylate/carbonyl and phenolic/alcoholic oxygen donors, with concurrent ligand/ion exchange at these sites. Together with the reduced O–H intensity, these changes support surface complexation as the dominant binding mechanism at pH 5.

3.3 Effect of pH

The effect of solution pH on the adsorption of Cd(II) was examined in the pH range of 2–7. The adsorption capacity increased with rising pH, peaking at pH 5. At lower pH values, the high concentration of H^+ ions competes with Cd^{2+} ions for adsorption sites, leading to reduced adsorption



efficiency (Teng et al., 2020). As pH increases, this competition diminishes, allowing more Cd^{2+} ions to interact with the available functional groups on the biochar surface, such as hydroxyl, carboxyl, and carbonyl groups, as supported by FTIR analysis. The pH was not raised beyond this level to avoid the potential formation of cadmium hydroxide species, which could lead to removal via precipitation rather than adsorption. Therefore, pH 5 was chosen as the optimum condition to ensure that the observed removal was primarily due to adsorption processes.

Although determination of the point of zero charge could provide additional insight into surface charge characteristics, the experimentally observed pH-dependent adsorption behavior, together with FTIR analysis, provides sufficient evidence to explain the adsorption mechanism in the present study.

TABLE 1 Different isotherm parameters for Cd(II) adsorption.

Isotherm parameters	MPB	MAB
Experimental value		
q_e (mg/g)	11.088	23.431
Freundlich isotherm		
K_f (mg/g)	2.728	12.971
n	1.972	2.865
R^2	0.984	0.854
Langmuir isotherm		
$q_{\max}$ (mg/g)	15.151	24.39
b (L/mg)	0.157	1.782
R^2	0.983	0.990
Sips isotherm		
q_m (mg/g)	17.858	24.195
K_{LF} (L/mg)	0.104	2.018
n	0.863	0.891
R^2	0.984	0.959
Fowler-guggenheim isotherm		
K_{FG} (L/mg)	2.67×10^{-4}	3.96×10^{-11}
W (kJ/mol)	-15.554	-36.884
R^2	0.992	0.935
Harkins-jura isotherm		
A	13.698	58.823
B	1.129	0.647
R^2	0.887	0.543

3.4 Effect of initial metal concentration

The results show that the efficiency of removing Cd(II) using MPB and MAB is affected by the initial Cd(II) concentration in the solution. The percentage of Cd(II) adsorption decreases as the initial concentration of Cd(II) increases, but this decrease is minimal at higher concentrations. This trend can be attributed to the limited number of available adsorption sites on the adsorbent surface, which gradually become saturated as the metal ion concentration increases (Figure 3B).

3.5 Effect of time

As shown in Figure 3C, Cd(II) adsorption onto MPB and MAB proceeded rapidly during the initial stage, with a substantial fraction of uptake occurring within the first 30 min. This behavior is attributed to the availability of a large number of vacant adsorption sites at the beginning of the process. As adsorption progressed and the number of available sites decreased, the adsorption rate slowed, approaching equilibrium after approximately 360 min.

3.6 Adsorption isotherm studies

The capacity of MPB and MAB to take up Cd(II) at different concentrations (5–30 mg/L) on a fixed amount of adsorbent at pH 5.0 was assessed using various adsorption isotherm models.

The Freundlich isotherm model (Freundlich, 1907) assumes a heterogeneous energetic distribution of active sites and interactions between adsorbed molecules, and the linear form of this model is expressed as:

$$\log q_e = \log K_f + (1/n) \log C_e \quad (2)$$

where C_e (mg/L) is the equilibrium concentration, q_e (mg/g) is the amount of metal ions adsorbed per specified amount of adsorbent at equilibrium, K_f (mg/g) is the adsorption capacity, and ' n ' represents the adsorption intensity. A plot of $\log q_e$ versus $\log C_e$ resulted in straight lines with correlation coefficients (R^2) of 0.984 and 0.854 for MPB and MAB, respectively (Figure 4A). The slope and intercept of the graph were used to calculate the values of n and K_f . The n values, which indicate the favorability of the adsorption process, were found to be 1.972 and 2.865, and K_f values were 2.728 and 12.971 mg/g for MPB and MAB, respectively (Table 1). An adsorption process is considered favorable if the value of n lies between 0 and 10 (Ahmadi et al., 2020).

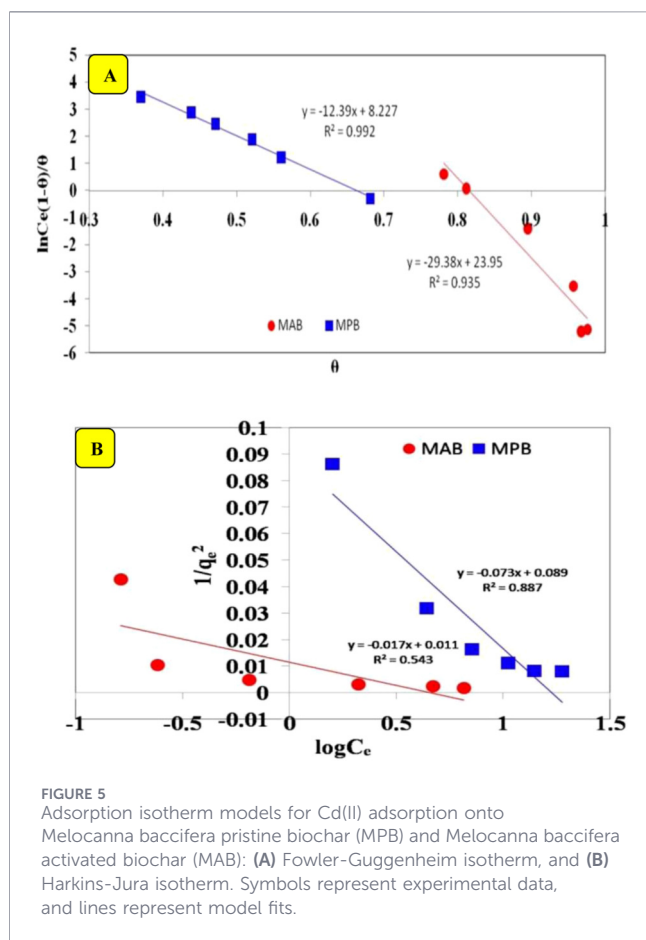
The Langmuir isotherm model (Langmuir, 1916) assumes that the adsorbent surface has a finite number of energetically uniform sites, and there is no interaction between adsorbed species once equilibrium is reached, forming a monolayer. The linear form of this model is given by:

$$C_e/q_e = 1/(q_{\max} b) + C_e/q_{\max} \quad (3)$$

where C_e (mg/L) is the equilibrium concentration of Cd(II), $q_{\max}$ (mg solute/g adsorbent) is the maximum adsorption capacity, and b (L/mg) is related to the energy of adsorption. Plotting C_e/q_e against C_e resulted in straight lines with correlation coefficients of 0.983 and 0.990 for MPB and MAB, respectively (Figure 4B). The $q_{\max}$ values were 15.151 mg/g for MPB and 24.39 mg/g for MAB, which correlated well with the theoretical $q_{\max}$ values (Table 1), suggesting adsorption behavior consistent with the Langmuir model.

The maximum adsorption capacity of Cd(II) achieved in this study using MAB was 24.39 mg/g. In comparison, other alkaline-activated adsorbents for Cd(II) adsorption reported in the literature show a wide range of capacities. For instance, biochar derived from *Enteromorpha prolifera* exhibits a adsorption capacity of 5.84 mg/g (Wang et al., 2021), while pyrolytic carbon has an adsorption capacity of 21.7 mg/g (Anum et al., 2025). Additionally, activated rice straw biochar demonstrates an adsorption capacity of 24.7 mg/g (Bashir et al., 2018). These values underscore the variability in Cd(II) uptake, which depends on the precursor material and the activation conditions.

Although the BET surface area increased by approximately 34% after activation, the adsorption capacity increased by about 60%. This disproportionate increase suggests that, in addition to surface area, the introduction of oxygen-containing functional groups and improved pore accessibility played a significant role in enhancing Cd(II) adsorption.



The Sips isotherm model, which combines features of the Langmuir and Freundlich models, and the non-linear form of this model is expressed as (de Vargas Briao et al., 2023):

$$q_e = q_m K_{LF} C_e^n / (1 + K_{LF} C_e^n) \quad (4)$$

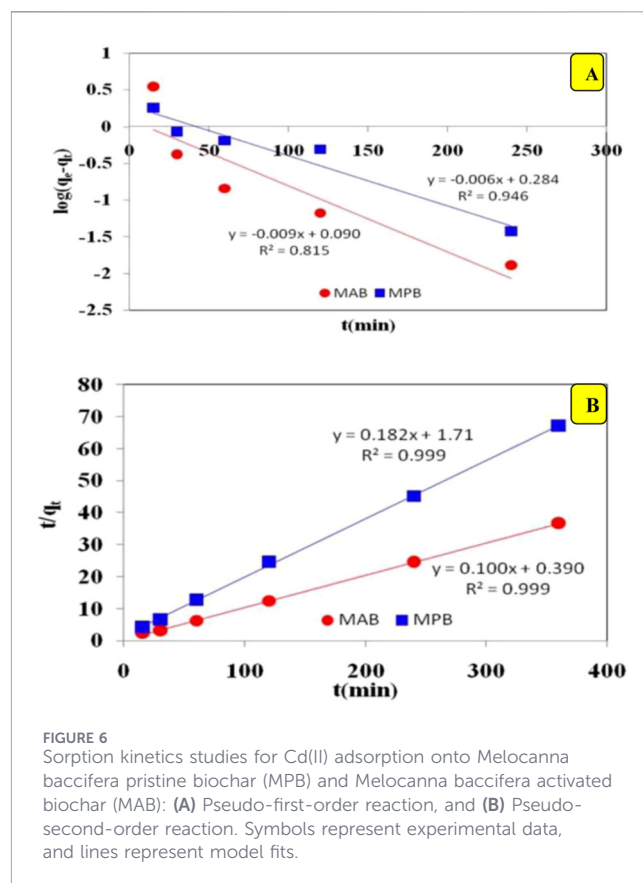
where C_e (mg/L) is the equilibrium concentration, q_m (mg/g) and K_{LF} (L/mg) are the Sips maximum adsorption capacity and Sips equilibrium constant, respectively, and n is the heterogeneity index. Values of $0 < n < 1$ indicate surface heterogeneity, whereas values approaching 1 suggest adsorption behavior similar to the Langmuir model (Chu et al., 2022).

Non-linear fitting of the Sips model (Figure 4C) yielded n values of 0.863 and 0.891 for MPB and MAB, respectively. These values, being close to unity, indicate adsorption behavior approaching Langmuir-type characteristics.

The goodness of fit for the non-linear Sips model was evaluated using the coefficient of determination (R^2), calculated from the sum of squared residuals (SSR) and total sum of squares (SST) following the method of Foo and Hameed (2010). The high R^2 values of 0.959 and 0.984 for MAB and MPB, respectively, indicate good agreement between the model predictions and experimental data.

The Fowler-Guggenheim isotherm (Fowler and Guggenheim, 1939) examines lateral interactions between adsorbed molecules in the solid phase. Its linear form is:

$$\ln C_e (1 - \theta) / \theta = -\ln K_{FG} + 2W\theta / RT \quad (5)$$



where K_{FG} (L/mg) is the Fowler-Guggenheim equilibrium constant, $\theta = (1 - C_e/C_i)$ is the degree of surface coverage, W (kJ/mol) is the interaction energy between adsorbed molecules, R is the universal gas constant and T (K) is the absolute temperature. Plotting $\ln C_e (1 - \theta) / \theta$ against θ allows for the calculation of W and K_{FG} (Figure 5A). For MPB and MAB, the ' W ' values were -15.554 kJ/mol and -36.884 kJ/mol, respectively, and the K_{FG} values are 2.67×10^{-4} L/mg and 3.96×10^{-11} L/mg with R^2 of 0.992 and 0.935, respectively. According to the model, positive ' W ' values indicate attractive interactions, while negative values suggest repulsive interactions among adsorbed molecules. Here, the negative ' W ' values suggest weak lateral repulsive interactions, and the high R^2 values indicate that the model adequately describes the data (Table 1). The negative ' W ' values derived from this model indicate weak lateral repulsion between adsorbed ions, which likely occurs at higher surface coverage. These interactions are not strong enough to create a significant deviation from Langmuir behavior, suggesting that the surface is largely homogeneous with only minor inter-adsorbate effects. The large difference in K_{FG} values between MPB and MAB reflects the substantial modification of surface chemistry and porosity after activation.

The Harkins-Jura isotherm model assumes the possibility of multilayer adsorption on adsorbents with heterogeneous pore distribution (Harkins and Jura, 1944). The linear form of the model is expressed as:

$$1/q_e^2 = B/A - (1/A) \log C_e \quad (6)$$

TABLE 2 Different kinetic models for Cd(II) adsorption.

Kinetic parameters	MPB	MAB
Experimental value		
q_e (mg/g)	5.353	9.794
Pseudo-first-order		
$q_{e,cal}$ (mg/g)	1.923	1.23
k_1 (/min)	13.81×10^{-3}	20.72×10^{-3}
R^2	0.946	0.815
Pseudo-second-order		
$q_{e,cal}$ (mg/g)	5.494	10.0
k_2 (g/mg/min)	19.37×10^{-3}	25.64×10^{-3}
R^2	0.999	0.999
Intraparticle diffusion		
k_{i1} (mg/g/min ^{0.5})	0.852	1.678
k_{i2} (mg/g/min ^{0.5})	0.062	0.012
C	4.224	9.57
R^2	0.935	0.91
Liquid film diffusion		
K_{LF} (cm/s)	0.015	0.02
R^2	0.946	0.815

The parameters were estimated using the linearized form of the Harkins–Jura equation. Plotting $1/q_e^2$ versus $\log C_e$ gave the R^2 of 0.887 for MPB and 0.543 for MAB (Figure 5B). The constants A and B were found to be 13.698 and 58.823, and 1.129 and 0.647 for MPB and MAB, respectively (Table 1). The lower R^2 values compared to the Langmuir model indicate that the Harkins–Jura model shows poor agreement, suggesting that MPB and MAB Cd(II) adsorptions are not well described by a multilayer adsorption process. The observed deviation from linearity, despite the use of linearized equations, reflects the poor applicability of the model rather than an issue with the fitting procedure.

3.7 Adsorption kinetics studies

To elucidate the adsorption kinetics and possible rate-controlling steps in the adsorption of Cd(II) onto MPB and MAB, various models were employed, including pseudo-first-order, pseudo-second-order, intraparticle diffusion, and liquid film diffusion models.

The pseudo-first-order model (Lagergren, 1898) suggests that the adsorption rate is proportional to the number of unoccupied sites. Its linear form is expressed as:

$$\log(q_e - q_t) = \log q_e - k_1 t / 2.303 \quad (7)$$

where q_e and q_t (mg/g) represent the amounts of Cd(II) adsorbed at equilibrium and at any time t , respectively, and k_1 (/min) is the rate constant. By plotting $\log(q_e - q_t)$ versus t (Figure 6A), the rate

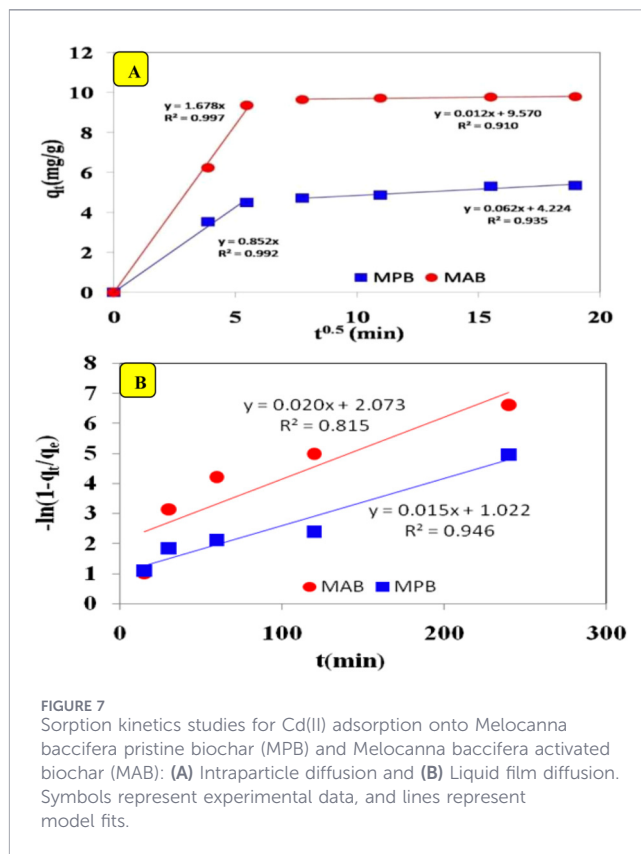


FIGURE 7

Sorption kinetics studies for Cd(II) adsorption onto Melocanna baccifera pristine biochar (MPB) and Melocanna baccifera activated biochar (MAB): (A) Intraparticle diffusion and (B) Liquid film diffusion. Symbols represent experimental data, and lines represent model fits.

constant k_1 and equilibrium adsorption capacity q_e were calculated (Table 2).

The pseudo-second-order (Ho and McKay, 1999) assumes that the rate of adsorption is proportional to the square of the number of unoccupied sites, and the linear form of the model is expressed as:

$$t/q_t = 1/k_2 q_e^2 + t/q_e \quad (8)$$

Where k_2 (g/mg/min) is the equilibrium rate constant, the plots of t/q_t versus t (Figure 6B) were used to determine the constants k_2 and q_e . Both the pseudo-first-order (PFO) and pseudo-second-order (PSO) models were assessed for comparative analysis. While the PFO model generated moderate R^2 values (0.946 for MPB and 0.815 for MAB), the predicted equilibrium adsorption capacities ($q_{e,cal}$) were significantly lower than the experimental q_e values, indicating a lack of suitability for the model. This inconsistency arises because the PFO model mainly fits the initial adsorption phase, often resulting in moderate R^2 values despite substantial discrepancies in q_e predictions (Ho and McKay, 1999; Azizian, 2004). In contrast, the PSO model produced $q_{e,cal}$ values that closely matched the experimental q_e , along with higher R^2 values (>0.99). This suggests that adsorption kinetics are better described by the pseudo-second-order model.

The linearized form of the intraparticle diffusion model proposed by Weber and Morris (Weber and Morris, 1962) is described by:

$$q_t = k_i t^{0.5} + C \quad (9)$$

where k_i (mg/g/min^{0.5}) is the intraparticle rate constant, and C represents the boundary layer thickness. The plot of q_t versus $t^{0.5}$

(Figure 7A) suggests that adsorption proceeds in multiple stages. For MAB, the value of $k_1 = 1.678 \text{ mg/g/min}^{0.5}$ decreased to $k_2 = 0.012 \text{ mg/g/min}^{0.5}$ with $C = 9.57$. For MPB, $k_1 = 0.852 \text{ mg/g/min}^{0.5}$ decreases to $k_2 = 0.062 \text{ mg/g/min}^{0.5}$ with a high value of $C = 4.224$ (Table 2). This suggests that intraparticle diffusion is faster initially, but not the sole rate-controlling step due to boundary layer effects.

The liquid film diffusion model, also known as external mass transfer (Salam et al., 2014). The linear form the model is expressed as:

$$-\ln(1 - F) = k_{LF}t \quad (10)$$

where $F = q_t/q_e$ is the fractional attainment of the equilibrium, k_{LF} (cm/s) is the liquid film diffusion rate constant. The plot of $-\ln(1 - F)$ versus t (Figure 7B) showed a high correlation coefficient ($R^2 = 0.946$ for MPB and 0.815 for MAB), suggesting that the adsorption process is also controlled by diffusion through the liquid film to the solid phase. The k_{LF} values were 0.015 cm/s for MPB and 0.02 cm/s for MAB (Table 2).

4 Conclusion

The study investigated the effectiveness of KOH chemical activation in improving biochar's ability to remove Cd(II) ions compared to acid activation. The results showed that the maximum adsorption capacities for Cd(II) were 15.15 mg/g for MPB and 24.39 mg/g for MAB at a pH of 5. The adsorption kinetics followed a pseudo-second-order model, indicating that the adsorption of Cd(II) involves interactions between two species. Moreover, the adsorption process is influenced by both intraparticle and liquid film diffusion mechanisms. Several isotherm models were used to describe the adsorption behavior. FTIR analysis of MPB and MAB showed the involvement of multiple functional groups in the adsorption of Cd(II) ions. Although the results demonstrate that activated bamboo biochar is an effective adsorbent for Cd(II) under laboratory conditions, several limitations should be acknowledged. Future studies should emphasize the limitations, conduct investigations with continuous-flow columns, and evaluate the application of *Melocanna baccifera* biochar using real wastewater. These investigations will enhance our understanding of the potential of this biochar for water treatment applications.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

HL: Methodology, Conceptualization, Supervision, Writing – review and editing, Writing – original draft,

Formal Analysis, Investigation. KL: Investigation, Writing – review and editing, Writing – original draft. LK: Writing – original draft, Conceptualization, Writing – review and editing. TS: Writing – original draft, Writing – review and editing, Conceptualization. LH: Formal Analysis, Writing – review and editing, Writing – original draft, Supervision.

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Conflict of interest

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/frcrb.2026.1720673/full#supplementary-material>

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